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Tuesday, October 6, 2026 11:51 AM

Showing weak stationarity

d) Show mean does not depend on t

$E[X_t] = E[Z_t Z_{t-1} \dots Z_{t-k}]$

$= E[Z_t]E[Z_{t-1}] \dots E[Z_{t-k}]$

$= 0 \cdot 0 \cdot \dots \cdot 0$

$= 0$

$\therefore$ constant $\forall t$

Show $\gamma(h)$ does not depend on t

$\gamma(h) = E[X_t X_{t+h}]$ (same reasoning as a, b, c)

$= E[(Z_t Z_{t-1} \dots Z_{t-k})(Z_{t+h} Z_{t-1+h} \dots Z_{t-k+h})]$

$= E[(\prod_{i=0}^{k} Z_{t-i})(\prod_{i=0}^{k} Z_{t-i+h})]$

There are two cases to take care of here.

Let $a = \prod_{i=0}^{k} Z_{t-i}$

Let $b = \prod_{i=0}^{k} Z_{t-i+h}$

Case 1: $h = 0$

If $h=0$, then a and b are in the same set perfectly.

This is shown as:

$a = \prod_{i=0}^{k} Z_{t-i}$

$b = \prod_{i=0}^{k} Z_{t-i+h}$

$= \prod_{i=0}^{k} Z_{t-i}$

$= a$ exactly.

Thus: $E[ab] = E[Z^2_t Z^2_{t-1} \dots Z^2_{t-k}]$

$= \sigma^2 \sigma^2 \dots \sigma^2$ $k+1$ factors since we start at $i=0$

$= \sigma^{2(k+1)}$

Case 2: $h \neq 0$

If $h \neq 0$, then b/c b will have at least one Z_{t-i+h} value offset from a.

Example: $h=1$

$a = \prod_{i=0}^{k} Z_{t-i}$

$b = \prod_{i=0}^{k} Z_{t-i+1}$

There is no Z_{t-i} factor in b, thus

$E[ab] = [Z_t Z^2_{t-1} \dots]$

(First factor of a; First factor of b)

$= 0 \cdot \sigma^2 \cdot \sigma^2 \dots$

$= 0$

This is true for any value of $h \neq 0$ b/c it is guaranteed to produce this offset.

$\therefore$ We have shown that

$$\gamma(h) = \begin{cases} \sigma^{2(k+1)} & \text{for } h = 0 \\ 0 & \text{fo } h \neq 0 \end{cases}$$

which does not depend on t.

$\therefore$ We have shown that X_t is weakly stationary. ■

Computing ACF

$$\gamma(h) = \begin{cases} \sigma^{2(k+1)} & \text{for } h = 0 \\ 0 & \text{for } h \neq 0 \end{cases}$$

$$\Rightarrow \rho(h) = \frac{\gamma(h)}{\gamma 0}$$

$$= \frac{0}{\sigma^{2(k+1)}}$$

$\therefore \rho(h) = 0$ is the ACF of X_t.